

# Malicious VBA

## Scenario:

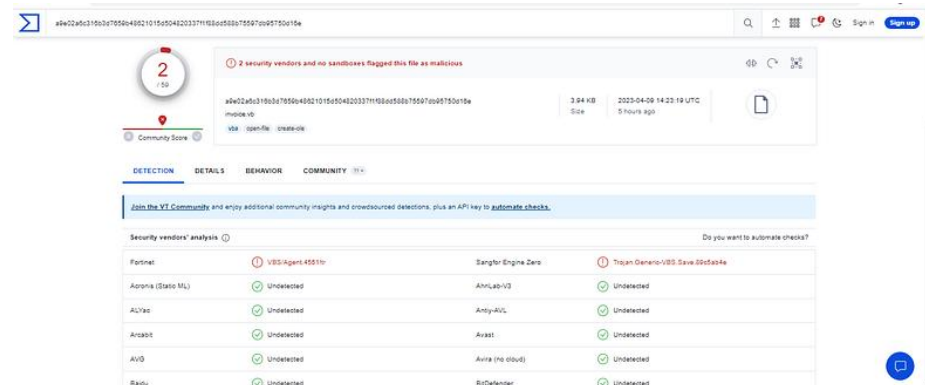
One of the employees has received a suspicious document attached in the invoice email. They sent you the file to investigate. You managed to extract some strings from the VBA Macro document. Can you refer to CyberChef and decode the suspicious strings?

## Objective:

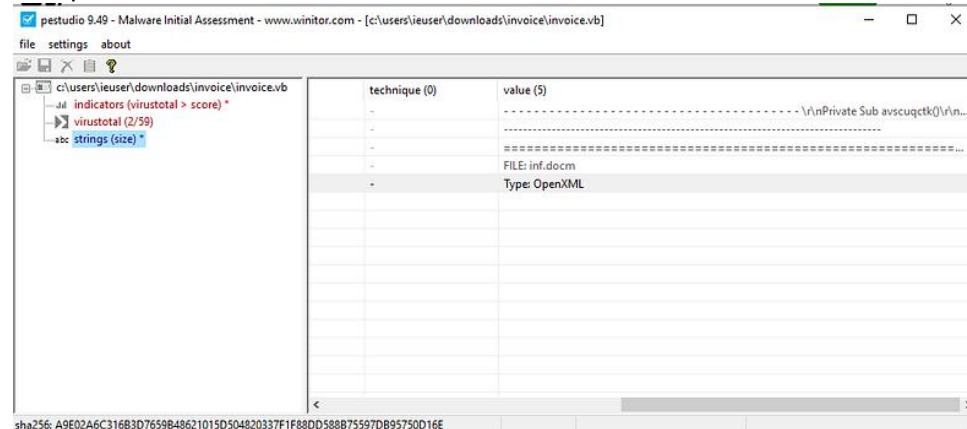
Analyze a suspicious document attached to an invoice email, extract and decode hidden strings from the VBA Macro, and determine if the file is malicious.

## Steps Taken:

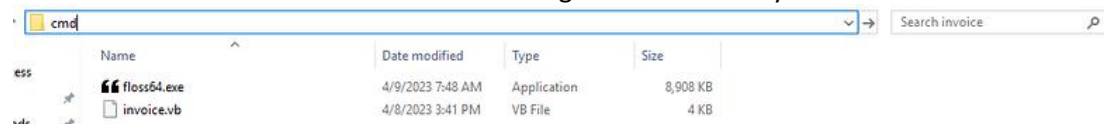
Searched the file on VirusTotal → only 2 vendors flagged it as malicious, but no detailed information was available.



Used PeStudio to inspect the file and extract initial strings, but results were too cluttered to interpret.



Switched to FLOSS tool to extract hidden strings more effectively:

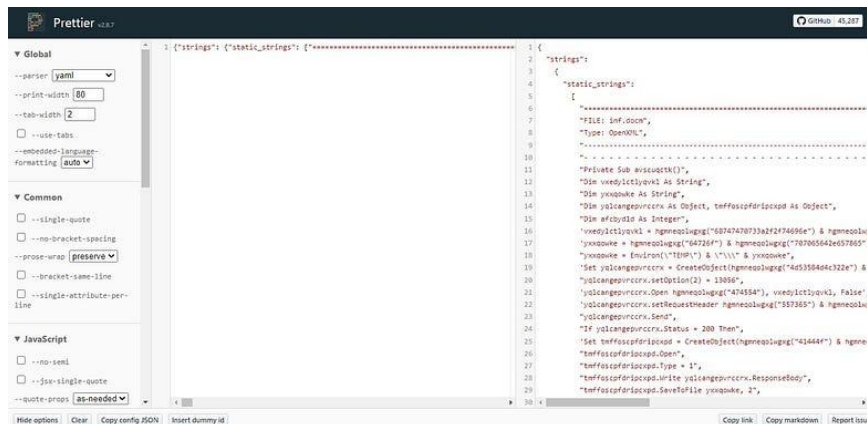


floss64.exe invoice.vb -o malfile.txt

→ Saved extracted strings into malfile.txt.

Observed that the code was obfuscated, making it difficult to understand in raw form.

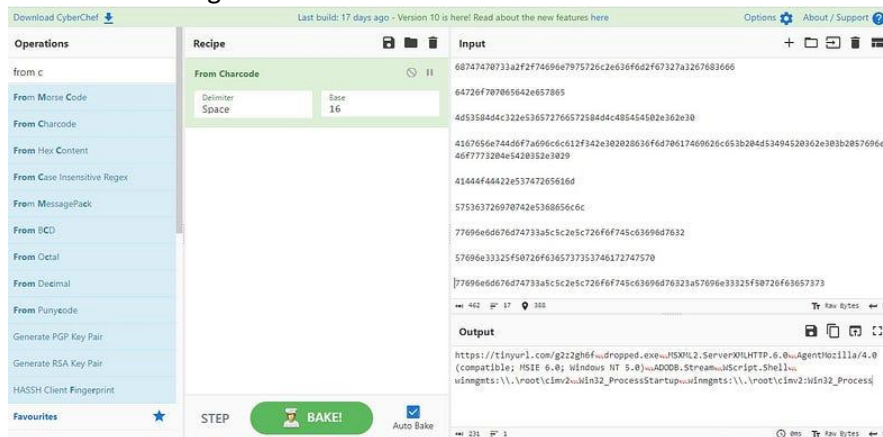
Reformatted the extracted code using Prettier (online formatter) for better readability.



Prepared two possible approaches to decode suspicious strings:  
 Identify the scripting language and analyze its behavior.  
 Extract Charcode values from the functions and decode them using CyberChef.

### Challenge 1:

The document initiates the download of a payload after the execution, can you tell what website is hosting it?



Using cyberchef website to decoded the code

**Answer :** <https://tinyurl.com/g2z2gh6f>

### Challenge 2:

What is the filename of the payload (include the extension)?

**Answer:** dropped.exe

### Challenge 3:

What method is it using to establish an HTTP connection between files on the malicious web server?

**Answer Q3:** MSXML2.ServerXMLHTTP

### Challenge 4:

What user-agent string is it using?

**Answer Q4:** Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.0)

### Challenge 5:

What object does the attacker use to be able to read or write text and binary files?

**Answer Q5:** ADODB.Stream

**Challenge 6:**

What is the object the attacker uses for WMI execution? Possibly they are using this to hide the suspicious application running in the background.

**Answer Q6:** winmgmts:\\.\root\cimv2:win32\_process